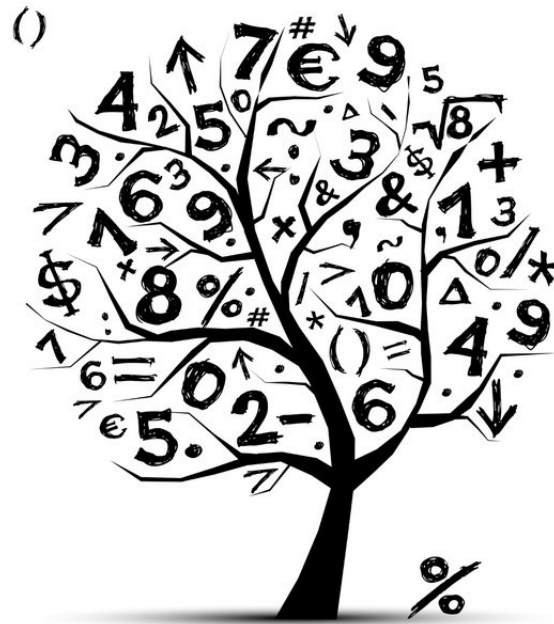


Factorization of matrices



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1 LU factorization

- The LU factorization (or LU decomposition) is a fundamental matrix decomposition that expresses a matrix as the product of a lower triangular matrix and an upper triangular matrix.

Given a square matrix $A \in \mathbb{R}^{n \times n}$, a **PLU factorization** of A is a decomposition of the form

$$PA = LU,$$

where:

- $P \in \mathbb{R}^{n \times n}$ is a **permutation matrix** (it accounts for possible row swaps),
- $L \in \mathbb{R}^{n \times n}$ is a **lower triangular matrix** with ones on the diagonal,
- $U \in \mathbb{R}^{n \times n}$ is an **upper triangular matrix**.

- The factorization is computed by Gaussian elimination.
- If no row swaps are needed, then $P = I_n$ and the factorization reduces to the **LU factorization**:

$$A = LU.$$

- The PLU/LU factorization is very useful for solving linear systems $Ax = b$ efficiently.

Method to compute the PLU (LU with pivoting) factorization.

Starting from A , we perform Gaussian elimination. Whenever a row swap is required, it is recorded in the permutation matrix P . At the end of the elimination we obtain the upper triangular matrix U , while L stores the elimination multipliers.

Specifically:

- Each elimination step $R_i \leftarrow R_i - \ell_{ij}R_j$ (with $i > j$) creates a zero in position (i, j) .
- The multiplier ℓ_{ij} is stored in position (i, j) of L .
- The diagonal of L is filled with ones.
- If at step j rows i and k are swapped ($R_i \leftrightarrow R_k$), the multipliers already stored in rows i and k of L (in columns $1, \dots, j-1$) are swapped as well, and the same swap is applied to the rows of P (which starts as I_n).

Example 1: LU factorization

Let us compute the LU factorization of

$$A = \begin{pmatrix} 2 & 1 & 1 \\ 4 & -6 & 0 \\ -2 & 7 & 2 \end{pmatrix}.$$

Step 1: Eliminate the entries below the first pivot $a_{11} = 2$.

We need to eliminate $a_{21} = 4$. The multiplier is:

$$\ell_{21} = \frac{a_{21}}{a_{11}} = \frac{4}{2} = 2.$$

We perform $R_2 \leftarrow R_2 - 2R_1$:

$$\begin{pmatrix} 2 & 1 & 1 \\ 0 & -8 & -2 \\ -2 & 7 & 2 \end{pmatrix}.$$

We need to eliminate $a_{31} = -2$. The multiplier is:

$$\ell_{31} = \frac{a_{31}}{a_{11}} = \frac{-2}{2} = -1.$$

We perform $R_3 \leftarrow R_3 - (-1)R_1 = R_3 + R_1$:

$$\begin{pmatrix} 2 & 1 & 1 \\ 0 & -8 & -2 \\ 0 & 8 & 3 \end{pmatrix}.$$

Step 2: Eliminate the entries below the second pivot -8 .

We need to eliminate the entry in position $(3, 2)$, which is currently 8. The multiplier is:

$$\ell_{32} = \frac{8}{-8} = -1.$$

We perform $R_3 \leftarrow R_3 - (-1)R_2 = R_3 + R_2$:

$$U = \begin{pmatrix} 2 & 1 & 1 \\ 0 & -8 & -2 \\ 0 & 0 & 1 \end{pmatrix}.$$

Example 2: LU factorization (continued)

Step 3: Build the matrix L .

The matrix L has ones on the diagonal and the multipliers below the diagonal:

$$L = \begin{pmatrix} 1 & 0 & 0 \\ \ell_{21} & 1 & 0 \\ \ell_{31} & \ell_{32} & 1 \end{pmatrix} = \begin{pmatrix} 1 & 0 & 0 \\ 2 & 1 & 0 \\ -1 & -1 & 1 \end{pmatrix}.$$

Verification: We check that $LU = A$:

$$\begin{aligned}
 LU &= \begin{pmatrix} 1 & 0 & 0 \\ 2 & 1 & 0 \\ -1 & -1 & 1 \end{pmatrix} \begin{pmatrix} 2 & 1 & 1 \\ 0 & -8 & -2 \\ 0 & 0 & 1 \end{pmatrix} \\
 &= \begin{pmatrix} 2 & 1 & 1 \\ 4 & 2-8 & 2-2 \\ -2 & -1+8 & -1+2+1 \end{pmatrix} \\
 &= \begin{pmatrix} 2 & 1 & 1 \\ 4 & -6 & 0 \\ -2 & 7 & 2 \end{pmatrix} = A.
 \end{aligned}$$

- Given $A \in \mathbb{R}^{n \times n}$ and $k \in \{1, \dots, n\}$, we denote by $A_{[k]}$ the **leading principal submatrix** of order k , i.e., the $k \times k$ matrix formed by the first k rows and the first k columns of A . Its determinant $\det(A_{[k]})$ is called the **leading principal minor** of order k . Note that $A_{[n]} = A$.

Observation 1: Existence of the LU factorization without row swaps

Let $A \in \mathbb{R}^{n \times n}$ be a nonsingular matrix. Then A admits a factorization

$$A = LU,$$

with L lower triangular with ones on the diagonal and U upper triangular, if and only if all its leading principal minors are nonzero:

$$\det(A_{[k]}) \neq 0, \quad \forall k \in \{1, \dots, n\}.$$

In this case, Gaussian elimination never requires a row swap and the pivots are

$$u_{11} = \det(A_{[1]}), \quad u_{kk} = \frac{\det(A_{[k]})}{\det(A_{[k-1]})}, \quad k \in \{2, \dots, n\}.$$

Proof. ($\Leftarrow$) We perform Gaussian elimination without row swaps. A row addition $R_i \leftarrow R_i - \ell_{ij}R_j$ with $j < i$ does not change any leading principal minor: if $k \geq i$, it acts on $A_{[k]}$ as a row addition; if $k < i$, it does not modify $A_{[k]}$. Suppose that the first $k-1$ steps have been performed. Then the leading principal submatrix of order k of the current matrix is upper triangular with diagonal entries $u_{11}, \dots, u_{kk}$, hence

$$\det(A_{[k]}) = u_{11} u_{22} \cdots u_{kk}.$$

Since $\det(A_{[k]}) \neq 0$, the pivot $u_{kk} = \det(A_{[k]}) / \det(A_{[k-1]})$ is nonzero and step k can be performed without a row swap.

($\Rightarrow$) If $A = LU$, since L is lower triangular and U is upper triangular, the first k rows and columns of the product only involve the first k rows and columns of the factors: $A_{[k]} = L_{[k]} U_{[k]}$. Hence $\det(A_{[k]}) = 1 \cdot u_{11} \cdots u_{kk}$. Since $\det(A) = u_{11} \cdots u_{nn} \neq 0$, all u_{ii} are nonzero, and so are all the leading principal minors. $\square$

- For the matrix of Example 1, we have $\det(A_{[1]}) = 2$, $\det(A_{[2]}) = 2 \cdot (-6) - 1 \cdot 4 = -16$ and $\det(A_{[3]}) = \det(A) = -16$, which give the pivots $u_{11} = 2$, $u_{22} = -16/2 = -8$ and $u_{33} = -16/(-16) = 1$.
- If some leading principal minor is zero, a row swap is needed and we must compute a PLU factorization (with $P \neq I$). This happens in the next examples.

Example 3: PLU factorization with row permutation

Consider the matrix

$$A = \begin{pmatrix} 0 & 1 \\ 1 & 1 \end{pmatrix}.$$

Since $a_{11} = 0$, we cannot use it as a pivot. We need to swap rows 1 and 2.

Let

$$P = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$$

be the permutation matrix that swaps rows 1 and 2.

Then:

$$PA = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} 0 & 1 \\ 1 & 1 \end{pmatrix} = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}.$$

Now PA already has an upper triangular form (no elimination needed below the diagonal), so:

$$L = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} = I, \quad U = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}.$$

Therefore, the PLU factorization is:

$$PA = LU,$$

where $P = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$, $L = I$, and $U = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$.

Example 4: PLU factorization with a row swap during the elimination

Let us compute the PLU factorization of

$$A = \begin{pmatrix} 2 & 1 & 1 \\ 4 & 2 & 3 \\ -2 & 3 & 1 \end{pmatrix}.$$

Note that $\det(A_{[2]}) = 2 \cdot 2 - 1 \cdot 4 = 0$: a row swap will be needed.

Step 1: Eliminate the entries below the first pivot $a_{11} = 2$. The multipliers are $\ell_{21} = \frac{4}{2} = 2$ and $\ell_{31} = \frac{-2}{2} = -1$:

$$\begin{pmatrix} 2 & 1 & 1 \\ 0 & 0 & 1 \\ -2 & 3 & 1 \end{pmatrix} \quad (R_2 \leftarrow R_2 - 2R_1)$$

$$\begin{pmatrix} 2 & 1 & 1 \\ 0 & 0 & 1 \\ 0 & 4 & 2 \end{pmatrix} \quad (R_3 \leftarrow R_3 + R_1)$$

Step 2: The entry in position (2,2) is 0 and cannot be used as a pivot. We swap rows 2 and 3:

$$\begin{pmatrix} 2 & 1 & 1 \\ 0 & 4 & 2 \\ 0 & 0 & 1 \end{pmatrix} \quad (R_2 \leftrightarrow R_3)$$

The multipliers already computed for rows 2 and 3 are swapped as well: now $\ell_{21} = -1$ and $\ell_{31} = 2$. The entry below the new pivot 4 is already zero, so $\ell_{32} = 0$, and the elimination is complete:

$$U = \begin{pmatrix} 2 & 1 & 1 \\ 0 & 4 & 2 \\ 0 & 0 & 1 \end{pmatrix}.$$

Step 3: Build P and L . The matrix P is obtained from I_3 by swapping rows 2 and 3, and L contains the (swapped) multipliers:

$$P = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{pmatrix}, \quad L = \begin{pmatrix} 1 & 0 & 0 \\ -1 & 1 & 0 \\ 2 & 0 & 1 \end{pmatrix}.$$

Verification: We check that $PA = LU$:

$$PA = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{pmatrix} \begin{pmatrix} 2 & 1 & 1 \\ 4 & 2 & 3 \\ -2 & 3 & 1 \end{pmatrix} = \begin{pmatrix} 2 & 1 & 1 \\ -2 & 3 & 1 \\ 4 & 2 & 3 \end{pmatrix},$$

$$LU = \begin{pmatrix} 1 & 0 & 0 \\ -1 & 1 & 0 \\ 2 & 0 & 1 \end{pmatrix} \begin{pmatrix} 2 & 1 & 1 \\ 0 & 4 & 2 \\ 0 & 0 & 1 \end{pmatrix} = \begin{pmatrix} 2 & 1 & 1 \\ -2 & -1+4 & -1+2 \\ 4 & 2 & 2+1 \end{pmatrix} = \begin{pmatrix} 2 & 1 & 1 \\ -2 & 3 & 1 \\ 4 & 2 & 3 \end{pmatrix} = PA.$$

- Without swapping the multipliers in Step 2 we would get the wrong matrix $\tilde{\mathbf{L}}$ with $\tilde{\ell}_{21} = 2$ and $\tilde{\ell}_{31} = -1$, and $\tilde{\mathbf{L}}\mathbf{U} \neq \mathbf{P}\mathbf{A}$ (its second row would be $(4, 6, 4)$).

2 Determinant from the PLU factorization

Observation 2: Determinant from the PLU factorization

Let $\mathbf{P}\mathbf{A} = \mathbf{L}\mathbf{U}$ be a PLU factorization of $\mathbf{A} \in \mathbb{R}^{n \times n}$, where $\mathbf{P}$ is obtained from $\mathbf{I}_n$ with s row swaps. Then

$$\det(\mathbf{A}) = (-1)^s \prod_{i=1}^n u_{ii} = \pm \prod_{i=1}^n u_{ii}.$$

In particular, $\mathbf{A}$ is invertible if and only if all the diagonal entries of $\mathbf{U}$ are nonzero.

Proof. Each row swap changes the sign of the determinant, so $\det(\mathbf{P}) = (-1)^s \det(\mathbf{I}_n) = (-1)^s$. The matrices $\mathbf{L}$ and $\mathbf{U}$ are triangular, so $\det(\mathbf{L}) = 1$ and $\det(\mathbf{U}) = \prod_{i=1}^n u_{ii}$. From $\det(\mathbf{P}\mathbf{A}) = \det(\mathbf{L}\mathbf{U})$ and the product rule of determinants,

$$(-1)^s \det(\mathbf{A}) = \prod_{i=1}^n u_{ii},$$

and we multiply both sides by $(-1)^s$. □

Example 5: Determinant from the PLU factorization

- Example 1: no row swaps ($s = 0$), so $\det(\mathbf{A}) = 2 \cdot (-8) \cdot 1 = -16$.
- Example 3: one row swap ($s = 1$), so $\det(\mathbf{A}) = -(1 \cdot 1) = -1$. Indeed, $\det(\mathbf{A}) = 0 \cdot 1 - 1 \cdot 1 = -1$.
- Example 4: one row swap ($s = 1$), so $\det(\mathbf{A}) = -(2 \cdot 4 \cdot 1) = -8$. Indeed, by Laplace expansion along the first row, $\det(\mathbf{A}) = 2 \cdot (2 - 9) - 1 \cdot (4 + 6) + 1 \cdot (12 + 4) = -14 - 10 + 16 = -8$.